

Mabel Echols

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EDUCATION

Northeastern University — Boston, MA

Expected Graduation: Dec 2026

Master of Science, Robotics, Concentration in Mechanical Engineering — (GPA: 4.0/4.0)

Carnegie Mellon University — Pittsburgh, PA

Graduated: May 2025

Bachelor of Science, Materials Science and Engineering

EXPERIENCE

Undergraduate Researcher — McGaughey Lab, Carnegie Mellon University

May 2024 - Aug 2024

- Worked with post-doctoral student to research thermal properties of semiconductor alloys using Molecular Dynamics (MD) simulations
- Implemented novel calculation methods for interatomic forces and atomic masses within the LAMMPS MD code

Materials Simulation and Testing Intern — Markforged

May 2023 - Aug 2023

- Developed and executed plan for measuring physical properties for 3D printed parts
- Integrated physical data into existing simulation software
- Polished changes in order to prepare for a launch to existing customer base

PROJECTS

Modular Pill Dispenser

2025

- Designed and prototyped a dispenser capable of dispensing precise quantities of medication and pre-programmed times
- Created modular components for chaining of an arbitrary number of modules

Atomic Layer Deposition Chamber

2025

- Worked with the Hackerfab research group out of CMU to develop an open source ALD from off the shelf components
- Designed and constructed vacuum chamber and vacuum compatible heater system using a combination of machined and off the shelf parts

Automatic Grain Boundary Segmentation Tool

2025

- Researched and developed methodologies for recognizing and marking grain boundaries from SEM micrographs of metal and ceramic samples
- Programmed Python library implementing my methodology for other researchers

SKILLS

Software: Python, MATLAB, C++, C#, Java, Javascript, Onshape, Solidworks, Google Suite, Ansys

Hardware: 3D Printing, Soldering, Lathe, Manual Mill, Basic Wood/Metalworking Tools

Materials Characterization: Tensile/Compression Testing, SEM, EDS, Impact Testing, XRD

Professional: Technical Writing, Experiment Design, Project Management